J-space, Part 2: The Model Matrix

From lab result to publishable finding — living handout

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Campaign started 2026-07-27 — preregistered before data (`27c6d3c`)

1 Where Part 1 left off, and what Part 2 must decide

Part 1 (Lab 37) replicated the descriptive geometry of the Anthropic workspace paper on `Olmo-3-32B-Think` at ∼10× thinner capacity, found the paper's static causal dissociation *null* on both OLMo and `Qwen3.6-27B` under energy-matched instruments, established per-item frozen J-ablation as a control-clean cross-model causal handle on retrieved content (−2.9/−2.4 nats), and measured a foil-calibrated 46-step workspace-ahead-of-text lead. Part 2 adjudicates *why the dissociation is missing*:

	hypothesis	discriminator
H1	externalization (think-training moved the workspace into tokens)	Workstream A: matched-pretraining sibling `Olmo-3.1-32B-Instruct`, Qwen matrix completion, `gemma-4-31B-it`
H2	occupancy (live J-space ≈ the output stream itself)	Workstream D: output-occupancy index across the matrix
H3	training-lab specifics	unfalsifiable; shrunk by exhausting the rest
H5	residual instruments	B (frozen-logit, fit-size, Dolma corpus, dose/persistence selection), C (load-demanding tasks, paper's own sets)

Standards: $n \geq 60$ headline cells, two seeds on decisive grids, bootstrap CIs, BH-FDR across the matrix, preregistered directional predictions (`preregistration.md`).

2 Workstream A

2.1 A0 — does the J-dictionary survive post-training?

The part-1 Think lens, applied unchanged (donor J, recipient unembedding) to `Olmo-3.1-32B-Instruct` — same base model, different post-training. Preregistered gate: probes ≥15/21 and multihop pass@1 $\geq 0.85\times$ donor $= 0.2408$.

Verdict: **`TRANSFER_FAIL`, informative.** The readout basis is static across post-training (unembed row cosine median 0.9984; final-norm gain cosine 1.000), direct-fact probes transfer exactly (17/21 both), the shortlist transfers — what post-training blunted is specifically the *rank-1 sharpening*, the very component that was the J-lens's distinctive advantage on Think (+41% relative pass@1 there, +8.5% here). The prereg FAIL rule fires: fit the sibling's own 120-prompt lens. Recovery to ∼0.28 ⇒ mid-band J-map drift under post-training; no recovery ⇒ Instruct resolves the silent bridge more weakly — itself an H1-flavored datum.

*Draft campaign handout; grows a subsection per banked phase. Part-1 handout: `jspace/handout/olmo32b_jspace_handout.pdf`.

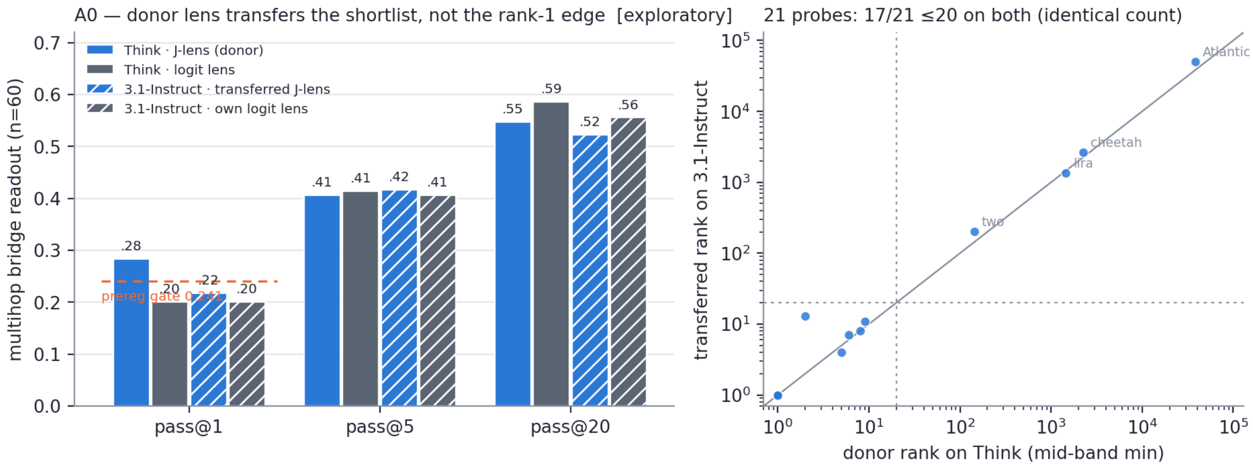

Figure 1: **A0 transfer gate.** Left: the transferred lens reads the bridge-entity *shortlist* at donor level (pass@5 0.417, pass@20 0.522 vs donor 0.41/0.55) but loses the rank-1 edge (0.217 vs 0.283, below the preregistered floor, dashed). Right: per-probe mid-band ranks — 17/21 ≤20 on *both* models, same items missing.

3 Workstream B

3.1 B3 — does the causal handle need the Jacobian? [exploratory pilot]

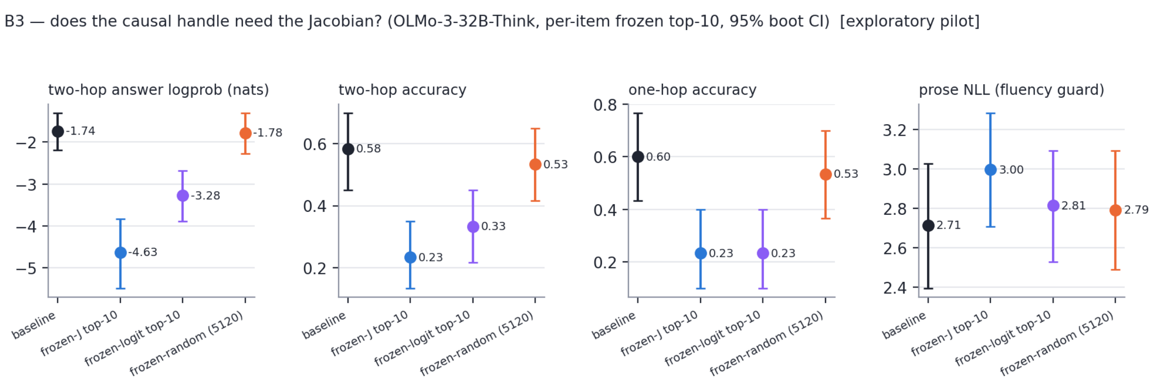

Figure 2: **B3.** The frozen-*logit* dictionary (no Jacobian, violet) reproduces ∼53% of frozen-J's two-hop logprob deletion (−1.54 vs −2.90 nats) and the *identical* one-hop collapse (0.23=0.23), while the random twin sits on baseline. Output-aligned directions alone carry much of the part-1 marquee effect; the geometry control family (R3) adjudicates the rest.

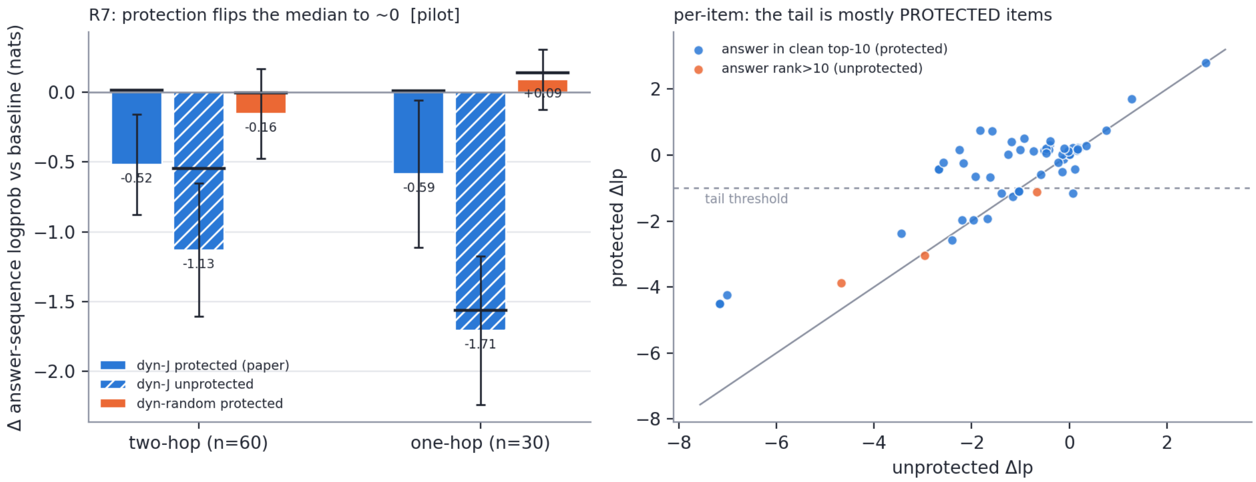

Figure 3: **R7 pilot (Think).** Left: with the paper's clean-output top-10 protection, the *median* item is untouched (two-hop +0.02, one-hop +0.01 nats) while the unprotected twin deletes hard (−0.55/−1.56 median) — part-1's live-ablation damage was mostly output-stream deletion (H2, causal form). Right: a J-specific tail survives protection (16/60 two-hop items lose >1 nat; protected-random on the same items: +0.34), and 13 of those 16 have the answer at clean rank 1–3 — *explicitly protected yet still deleted*: indirect J-content damage (bridge-entity reading), the confirmatory grid's prime target.

4 Workstream R — assay repair (the addendum's mandate)

4.1 R7 pilot — the paper's output-protected ablation, run faithfully

4.2 R2 — capacity under the paper's own estimator

4.3 R7 cross-model + R2-Qwen — the campaign headline (pilot)

4.4 The four-model ladder — the causal signature reorganizes

4.5 What the protected ablation actually deletes (tail mechanism)

The selected-id instrument logs, per layer and position, which dictionary rows the ablator deflates. On Olmo-3-32B-Think: the two-hop bridge entity's tokens are deflated on **95%** of items versus **2%** under a bridge-reassignment permutation ($p < 0.0005$) — the J-directions really do carry the intermediate entity. But this is *universal*: 0.94 on the damaged tail versus 0.95 on undamaged items, so bridge deletion alone is not sufficient for damage. What separates the tail is (i) protection failure — 21% of tail items had answers outside the protected clean-top-10 versus 0% of non-tail — and (ii) among the 19 tail items whose answer *was* protected, lower model confidence (median clean rank 2 versus 1). Deleting workspace content costs the model only when the answer is not already locked in.

4.6 Gemma-4-31B: an architecture datum with a depth

4.7 Design consequence: the confirmatory endpoint must change

The G6 power simulation, calibrated on the pilot parquets, finds that protected paired deltas are a zero-mode plus heavy-tail mixture (within-family σ 1.2–1.5 nats, ICC ≈ 0.40). The drafted 0.5-nat *mean* primaries reach only 24–42% power even at n=180 across 60 families, and no affordable n reaches 90%; equivalence testing at the same margin needs $n \approx 300$. The *tail-rate* endpoint — per-item >1-nat protected deletion, J versus matched random, paired and family-clustered — is well powered at a 10 pp margin. The preregistration records three options and awaits the principal investigator's choice; nothing downstream freezes until then.

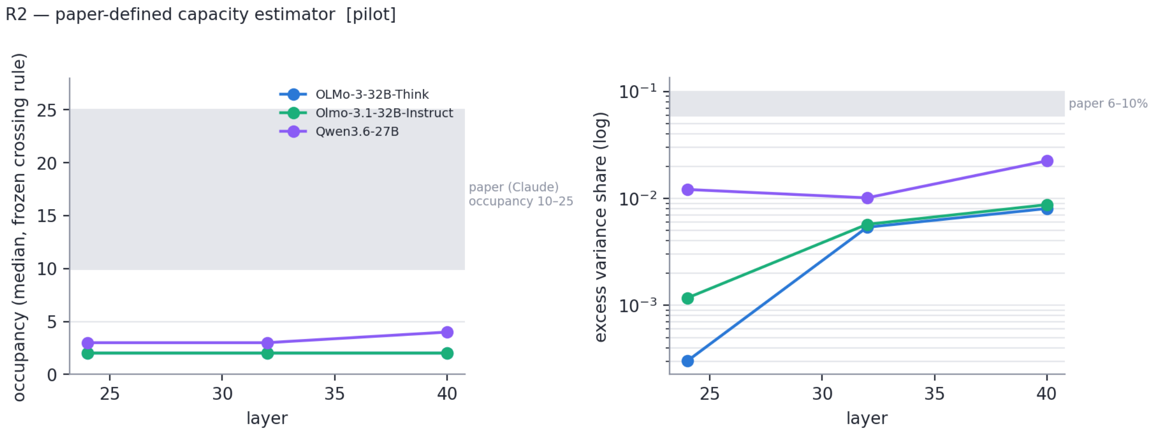

Figure 4: **R2 pilot.** Marginal-gain occupancy (frozen crossing rule, 3 random control dictionaries) and excess variance share at median occupancy. Both OLMo siblings: occupancy median 2, excess $\leq 0.9\%$ — far below the paper's Claude band (10–25, 6–10%), identical across the post-training pair. The Qwen recompute under this same estimator is the decisive cross-model capacity cell.

Reproducibility

Run dir `special-lab-1/part2_20260727/` (metrics per model slug, lenses, logs); code + results mirrored to `interpretability/jspace/part2/` on branch `interp_jspace_part2`; every figure regenerates from metrics via `scripts/p2fig.py`; master dataset `report/matrix_master.csv`.

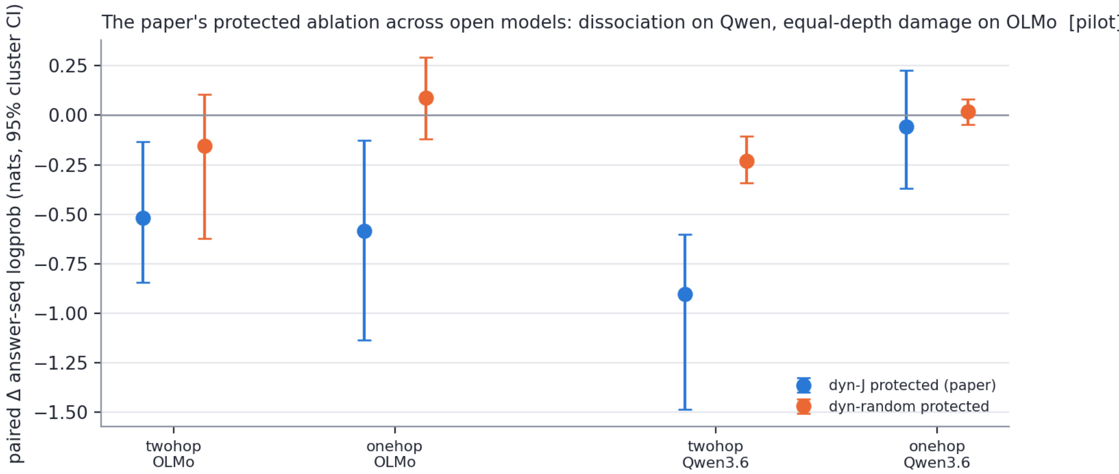

Figure 5: **Capacity and causal signature co-vary.** Under the identical protected protocol (paired family-clustered CIs): Qwen3.6-27B shows the paper's dissociation shape — two-hop −0.91 [−1.49, −0.60], one-hop −0.06 [straddles 0] — while OLMo-3-32B-Think shows equal-depth content damage (−0.52/−0.59, both excl. 0, plus the indirect protected tail). The same models order the same way on paper-defined occupancy (3–4 vs 2, excess 1.0–2.2% vs ≤0.9%), and BOTH sit an order of magnitude below the paper's Claude band (10–25, 6–10%) — with lens fit size ruled out (Qwen used the published n=1000 lens). Lens robustness: two independent 120-prompt Instruct fits agree at dictionary row-cos 0.993 and frozen-selection Jaccard 0.806.

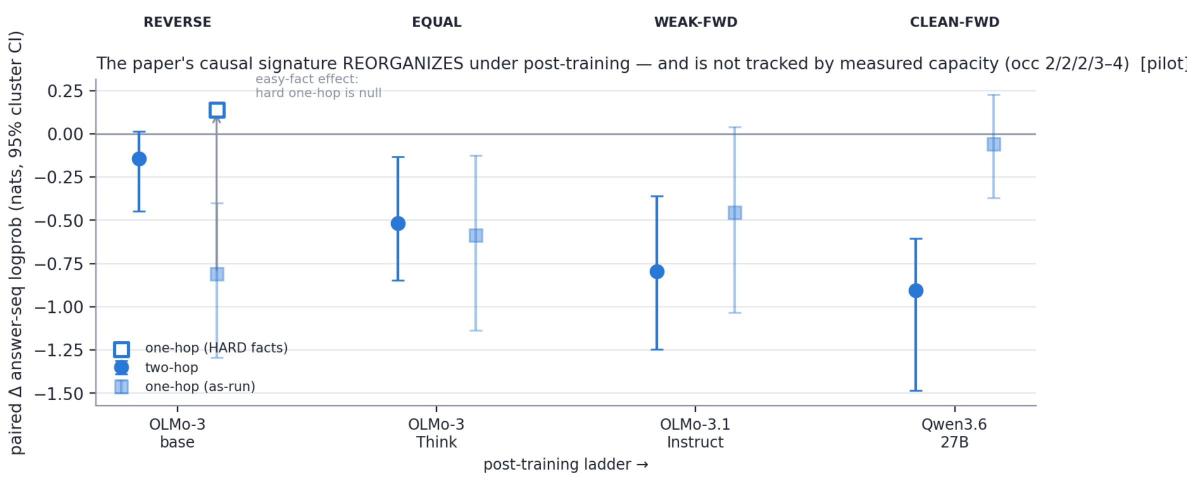

Figure 6: **The paper's causal signature is not a fixed property of "an LLM" — it reorganizes along post-training.** Protected dynamic J-ablation, paired family-clustered CIs, four checkpoints: the base model shows a REVERSE dissociation (one-hop hit, two-hop spared), Think shows equal-depth damage, 3.1-Instruct a weak forward dissociation, Qwen the paper's clean shape. Measured capacity does *not* track this (occupancy 2/2/2/3–4): Instruct and Think are identical on capacity and differ on shape. The open marker records the ceiling check that relabeled the bottom rung — on *hard* one-hop facts the base effect is null (+0.14; matched random −0.01), so the −0.81 was an EASY-FACT effect: rehearsed facts occupy the deletable output-adjacent channel and hard facts do not. Confirmatory hypotheses are therefore stated in fact-accessibility terms, not task depth.

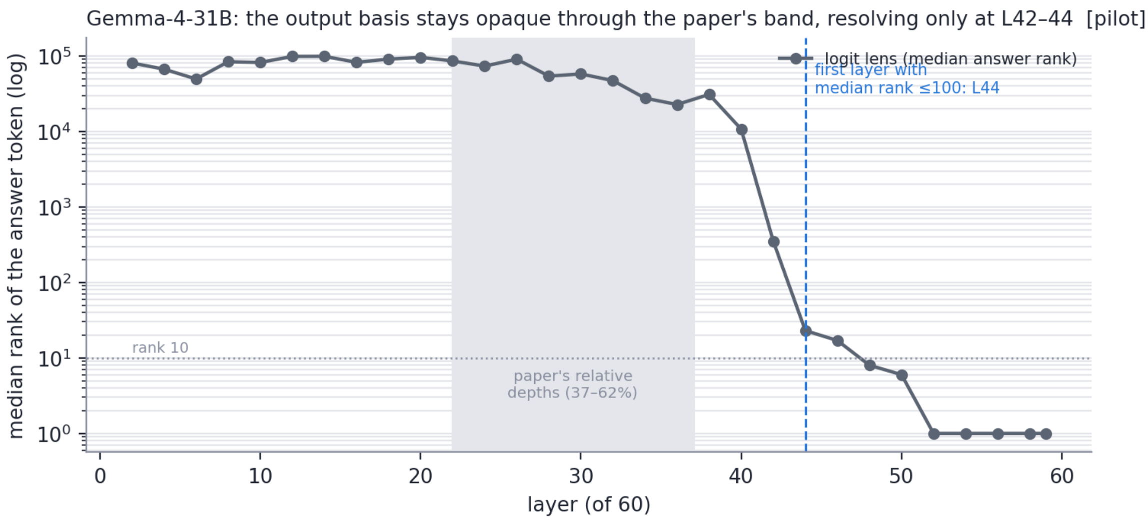

Figure 7: **Gemma's residual stream is opaque in the output basis through exactly the band the paper studies.** Median rank of the answer token by layer, read with the reference unembedding (final norm, tied head, softcap 30). Ranks stay in the tens of thousands from L2 to L40 and collapse abruptly at L42–44, reaching rank 1 by L52. The paper's relative depths (37–62%, i.e. L22–L37) sit entirely inside the opaque zone. The vanilla logit lens and the Jacobian micro-lens fail *equally* here, so this is a property of the model family, not of the instrument. Whether Jacobian transport rescues mid-band readability is the question the full 120-prompt fit answers; no family verdict is stated until it lands.